

House Price Prediction using Simple Linear Regression

Project Overview:

This project predicts house prices using a Machine Learning Linear Regression model.

The model learns relationships between housing features and predicts house prices.

Objective:

Build a Linear Regression model to predict housing prices based on different features.

Dataset:

California Housing Dataset from Scikit-learn.

Features:

- MedInc: Median income of residents
- HouseAge: Average age of houses
- AveRooms: Average number of rooms
- AveBedrms: Average bedrooms
- Population: Area population
- AveOccup: Average occupants per household
- Latitude: Location latitude
- Longitude: Location longitude

Technologies Used:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow:

1. Dataset Loading
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Selection
5. Feature Scaling
6. Train-Test Split
7. Linear Regression Model Training
8. Prediction
9. Model Evaluation

Machine Learning Model:

Algorithm: Simple Linear Regression

Evaluation Metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R2 Score

House Price Prediction:

Example input:

MedInc = 8.5

HouseAge = 25

AveRooms = 7

AveBedrms = 1

Population = 300

AveOccup = 2

Latitude = 40

Longitude = -120

Project Structure:

House-Price-Prediction/

- Housing_Price_Prediction.ipynb
- README.md
- requirements.txt

How to Run:

1. Install required libraries:

```
pip install -r requirements.txt
```

2. Open Jupyter Notebook:

```
jupyter notebook
```

3. Run all cells in Housing_Price_Prediction.ipynb

Requirements:

pandas

numpy

matplotlib

seaborn

scikit-learn

jupyter

Conclusion:

The Linear Regression model successfully predicts house prices using housing features.

This project demonstrates a complete Machine Learning workflow from preprocessing to prediction.

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Data Science / Machine Learning Project